

Name: Pramod Pandit

Reg no.: 12305901

1. Problem: Maximum Subarray Sum

```
2. int MaxSubarraySum(vector<int> arr) {
3.     int currSum = 0;
4.     int maxSum = INT_MIN;
5.
6.     for(int x : arr) {
7.         currSum += x;
8.
9.         maxSum = max(maxSum, currSum);
10.
11.         if(currSum < 0){
12.             currSum = 0;
13.         }
14.     }
15.     return maxSum;
16. }
```

2. Problem: Maximum Sum Subarray of Size K

```
17.
18. int MaxSum(vector<int> arr, int k) {
19.     int wSum = 0;
20.
21.     for(int i = 0; i < k; i++) {
22.         wSum += arr[i];
23.     }
24.
25.     int maxSum = wSum;
26.
27.     for(int i = k; i < arr.size(); i++) {
28.         wSum += arr[i];
29.         wSum -= arr[i - k];
30.
31.         maxSum = max(maxSum, wSum);
32.     }
33.     return maxSum;
34. }
```